

6 The kidney is involved in osmoregulation and the production of urine.

(a) (i) Which substances are filtered from the blood in the renal capsule?

(1)

- ☐ **A** glucose and fibrinogen
- ☐ **B** glucose, fibrinogen and urea
- ☐ **C** glucose and urea
- ☐ **D** urea and prothrombin

(ii) Which structure is acting as the filter in the renal capsule?

(1)

- ☐ **A** basement membrane
- ☐ **B** cells lining the renal capsule
- ☐ **C** endothelial cells in the walls of the capillaries
- ☐ **D** podocytes

(iii) Which transport mechanism is responsible for the uptake of glucose into the cells of the wall of the proximal tubule?

(1)

- ☐ **A** diffusion
- ☐ **B** endocytosis
- ☐ **C** sodium co-transport
- ☐ **D** osmosis

(iv) Explain why some urea is reabsorbed in the proximal tubule.

(2)

- (b) In an investigation, salt solution was injected into the carotid artery of a mammal. The carotid artery carries blood to the head.

The rate of urine production was measured after the injection.

The table shows the results.

Time after injection of salt solution / min	Rate of urine production / $\text{cm}^3 \text{ min}^{-1}$
0	7.0
5	1.1
10	0.6
15	1.0
20	1.5
25	2.2
30	2.4

- (i) Calculate the percentage decrease in the rate of urine production 10 minutes after the injection.

Give your answer to 2 decimal places.

(2)

Answer%

- (ii) At what times should more measurements be taken to find the lowest rate of urine production?

(1)

- ☐ **A** at 1-minute intervals between 5 minutes and 10 minutes
- ☐ **B** at 1-minute intervals between 5 minutes and 15 minutes
- ☐ **C** at 1-minute intervals between 10 minutes and 15 minutes
- ☐ **D** at 5 minutes, 10 minutes and 15 minutes, two more times

(iii) Explain the results of this investigation. Use the information in the table to support your answer.

(4)

[illegible]

(Total for Question 6 = 12 marks)